

EXPERIMENT – 01

Objective :- Implementation of BUS, STAR, MESH & Hybrid topology in CISCO Packet Tracer using switches and end devices.

BUS TOPOLOGY

Steps :

1. Open Cisco Packet Tracer.
2. From End Devices, drag and drop:
 - 3–4 PCs into the workspace.
3. From Network Devices → Switches, drag and drop:
 - 1 Switch (e.g., 2960).
4. Click on Connections (⚡ icon).
5. Select Copper Straight-Through Cable.
6. Connect:

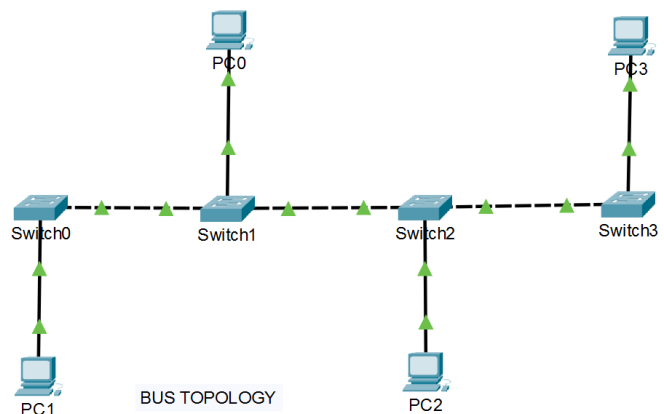
- PC0 → Switch
- PC1 → Switch
- PC2 → Switch
- PC3 → Switch

7. Configure IP Address on each PC:

- PC0 → 192.168.1.1
- PC1 → 192.168.1.2
- PC2 → 192.168.1.3
- PC3 → 192.168.1.4
- Subnet Mask → 255.255.255.0

8. Test connectivity:

- Desktop → Command Prompt
- Type: ping 192.168.1.2



STAR TOPOLOGY

Steps:

1. Open Cisco Packet Tracer.
2. From End Devices, drag and drop:
 - 4 PCs into the workspace.
3. From Network Devices → Switches, drag and drop:
 - 1 Switch (2960) in the center.
4. Click on Connections (🔗 icon).
5. Select Copper Straight-Through Cable.
6. Connect each PC to the Switch:

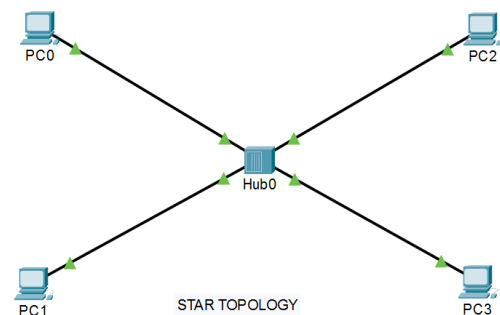
- PC0 → Hub
- PC1 → Hub
- PC2 → Hub
- PC3 → Hub

7. Configure IP Address on each PC:

- PC0 → 192.168.1.1
- PC1 → 192.168.1.2
- PC2 → 192.168.1.3
- PC3 → 192.168.1.4

8. Test the network:

- Desktop → Command Prompt
- Type: ping 192.168.1.2



HYBRID TOPOLOGY

Steps:

1. Open Cisco Packet Tracer.
2. From End Devices, drag and drop:
 - 4–6 PCs.
3. From Network Devices → Switches, drag and drop:
 - 2 Switches (2960).
4. Arrange like this:

- PC0, PC1 → Switch0
- PC2, PC3 → Switch1

5. Click on Connections (🔗 icon).

6. Select Copper Straight-Through Cable.

7. Connect:

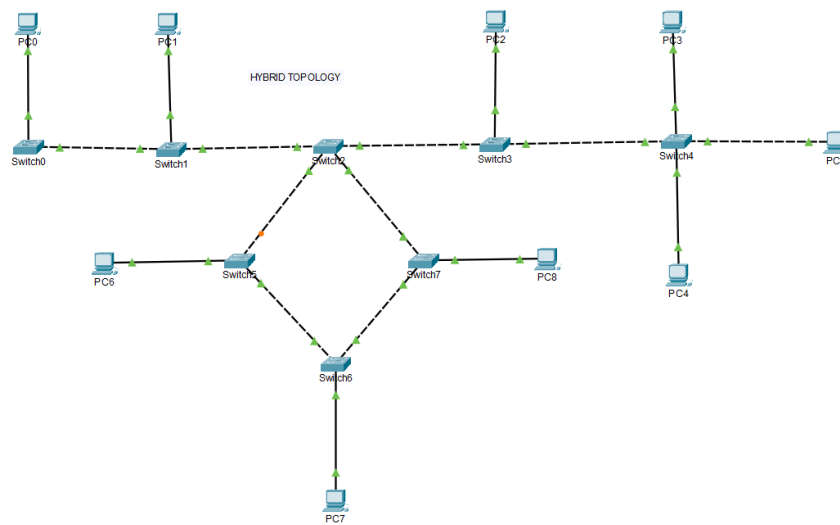
- PCs to their respective Switch
- Switch0 → Switch1

8. Configure IP Address:

- PC0 → 192.168.1.1
- PC1 → 192.168.1.2
- PC2 → 192.168.1.3
- PC3 → 192.168.1.4
- Subnet Mask → 255.255.255.0

9. Test connectivity:

- Desktop → Command Prompt
- Type: ping 192.168.1.3



MESH TOPOLOGY

Steps:

1. Open Cisco Packet Tracer.
2. From End Devices, drag and drop:
 - 3–4 PCs into the workspace.
3. Click on Connections (🔗 icon).
4. Select Copper Cross-Over Cable.
5. Connect each PC to every other PC:
 - PC0 → PC1
 - PC0 → PC2
 - PC1 → PC2
6. Configure IP Address:
 - PC0 → 192.168.1.1
 - PC1 → 192.168.1.2
 - PC2 → 192.168.1.3
 - Subnet Mask → 255.255.255.0
7. Test network:
 - Desktop → Command Prompt
 - Type: ping 192.168.1.2

